PROJECT OVERVIEW

Logbook 42·50

A training-plan and running-journal app that builds its own programme from race dates, reads watch data, and warns before injuries happen, for the road from a half marathon to a full marathon and a 50 km ultra.

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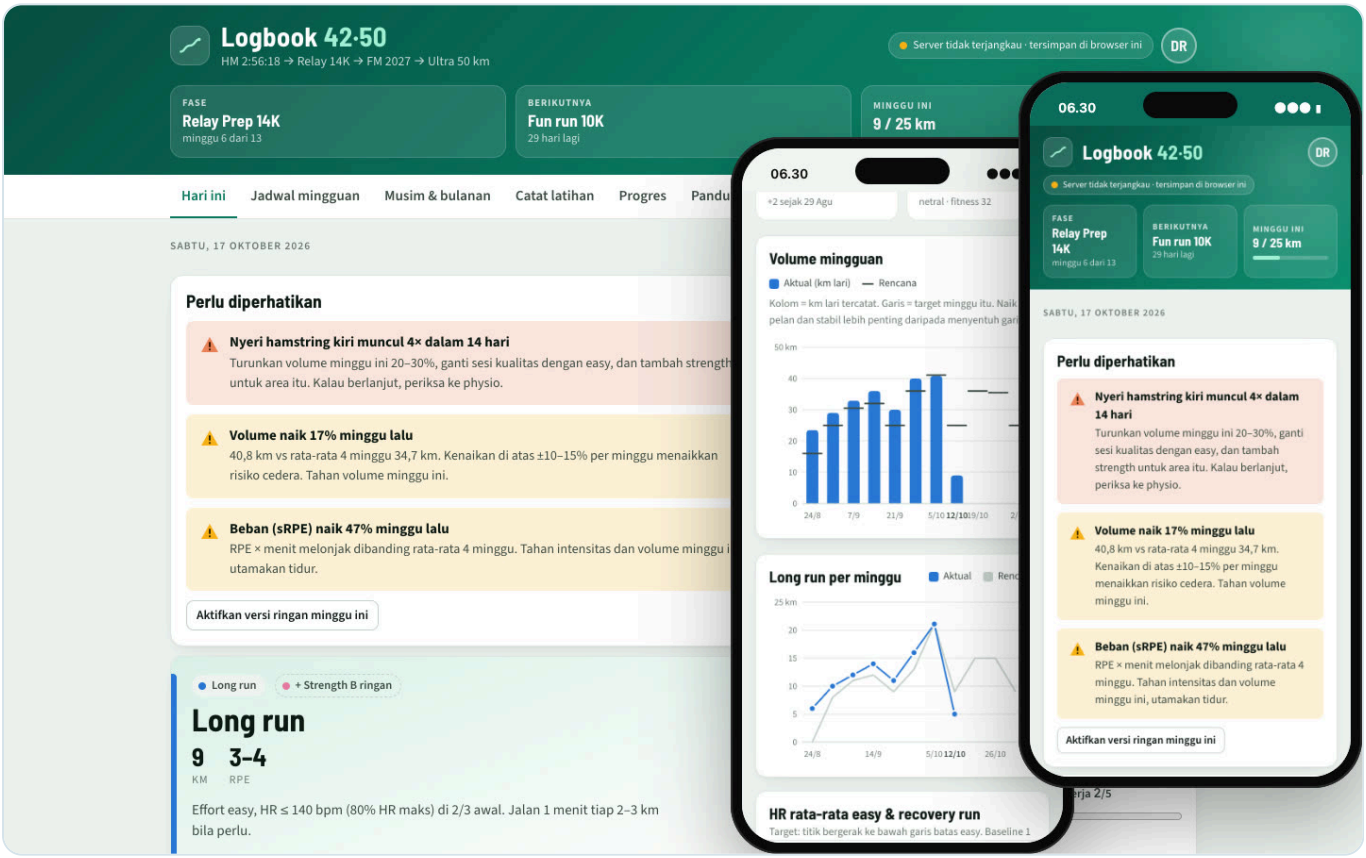
ROLE

Personal project · design, frontend, backend, deployment · used daily by me

PLATFORMS

PWA (iPhone, Android, desktop), installable on the Home Screen and usable offline

Vanilla JavaScript · PWA + Service Worker · Vercel Functions & Cron · Upstash Redis · Gemini API · intervals.icu API · Web Push · jsPDF · OpenStreetMap · Open-Meteo



OVERVIEW

Typical running apps record what already happened, but they don't build a programme toward several goals at once: a 14 km relay leg, a full marathon, mountain hikes, then a 50 km ultra. A spreadsheet plan also can't tell when sleep is short, pain keeps coming back in the same spot, or volume climbs too fast.

Logbook 42·50 generates a weekly plan from the race dates and adjusts it using a morning check-in, logged sessions, and watch data (Huawei via intervals.icu). It warns about training load, predicts race times, builds a race-day plan, and has an AI coach (Gemini) for daily, weekly, and monthly reviews.

It is a single index.html with no framework and no build step, plus 9 small Vercel Functions backed by Upstash Redis. It installs to the Home Screen, opens without signal, and syncs across devices. I use it every day; the portfolio video and screenshots use demo data.

KEY FEATURES

Plan generated from race dates

Training phases are built backwards from the relay, marathon, hike, and ultra dates. Each week has quality sessions, a long run, easy runs, and strength A/B. Changing a date or adding an event rebuilds the whole season, including adjustments for Ramadan and Monday/Thursday fasting.

Morning check-in & readiness

Sleep, stress, soreness, and pain (with a body map) are logged each morning; sleep, resting HR, and HRV from the watch fill in automatically. The result is advice for the day: run as planned, go lighter, or swap for a short easy run.

Three ways to log a session

The manual form works out pace and category from HR and pace. Huawei Health screenshots are read by Gemini (or Tesseract OCR in the browser without an API key). Watch sessions arrive by themselves through intervals.icu and merge with a similar manual entry instead of duplicating it.

Warnings before injuries

The app flags volume rising more than 15%, sRPE load spikes, pain recurring in the same area within two weeks, missed sessions, and shoes nearing their mileage limit, then offers a lighter version of the rest of the week.

Progress & race predictions

Charts for weekly volume, long runs, easy-run HR, VO2max, and Fitness · Fatigue · Form; achievements and streaks; Road to race; monthly recaps; sRPE load; and race-time predictions from the Riegel formula against targets.

Session detail from the watch

For synced sessions: a route map (OpenStreetMap), pace, HR, cadence, and elevation charts across the run, time in each HR zone, and per-km splits. A Share button renders an image card for WhatsApp or stories.

Race-day plan

For each race: pace per segment from the target time, a gel and drink schedule by minute, and an hourly fluid target from the sweat-rate calculator in the profile. The page can be printed or saved as PDF.

AI coach (Gemini)

Daily, weekly (automatic at the start of each week), and monthly reviews, plus free-form questions with the plan, the last four weeks of sessions, and check-ins as context. The prompt forbids making up data and points to a physiotherapist for pain of 4/10 or more.

Reminders, calendar & reports

Morning and evening push notifications via Vercel Cron, an .ics calendar feed with alarms for Apple/Google Calendar, a read-only coach link, a PDF report (profile, records, every session, per-session detail), plus JSON backup and CSV export.

ARCHITECTURE & DATA FLOW

PWA (index.html)
All UI and planning logic in the browser; local data copy + write queue while offline

→

Vercel Functions
/api/data, /api/doc, intervals, import, review, ask, push, calendar, coach

→

Upstash Redis
One hash per collection, one record per field (sessions, check-ins, weeks, events, reviews)

→

Layanan luar
intervals.icu (Huawei watch), Gemini, Web Push, Open-Meteo, OpenStreetMap

- The page loads everything once through GET /api/data, then sends each change as a single record through POST /api/doc. Other devices pull changes every 30 seconds, when the tab is reopened, and when the connection comes back.
- The service worker caches the app shell network-first, while /api is never cached. Writes made offline go into a localStorage queue and are sent automatically once online.

- Every API key (intervals.icu, Gemini, VAPID, Redis) lives only in Vercel env. The browser sends the app password in a header, and the server compares it with timingSafeEqual.

STACK BY LAYER

Frontend	HTML, CSS, JavaScript tanpa framework · Grafik SVG buatan sendiri · Tema terang & gelap
PWA	Service Worker (network-first) · Web App Manifest · Web Push (VAPID) · localStorage + antrean offline
Backend	Vercel Functions (Node.js) · Vercel Cron · web-push (satu-satunya dependency)
Data	Upstash Redis (REST pipeline) · Backup JSON · Ekspor CSV
Integrations	intervals.icu API · Gemini API (REST) · Tesseract.js (OCR cadangan) · Open-Meteo · OpenStreetMap
Documents	jsPDF (dimuat saat dibutuhkan) · Feed kalender .ics

TECHNICAL DECISIONS

1. **One file, no framework, no build**
   The whole app lives in index.html. Deploying is just a push, there are no frontend dependencies to update, and the service worker only needs to cache one page for the app to open offline. The trade-off is a large file, so the code is grouped by feature with marker comments.

2. **One record per Redis field**
   Each session, check-in, or event is its own field in a Redis hash and is written individually (64 KB max). Two devices editing different records never overwrite each other, unlike saving all the data as a single document.

3. **The plan is computed in the browser**
   The server only stores raw data; the plan, warnings, and predictions are recomputed in the browser, so they still work offline. For push notifications the page uploads the next few weeks of the plan so the cron job knows the day's session.

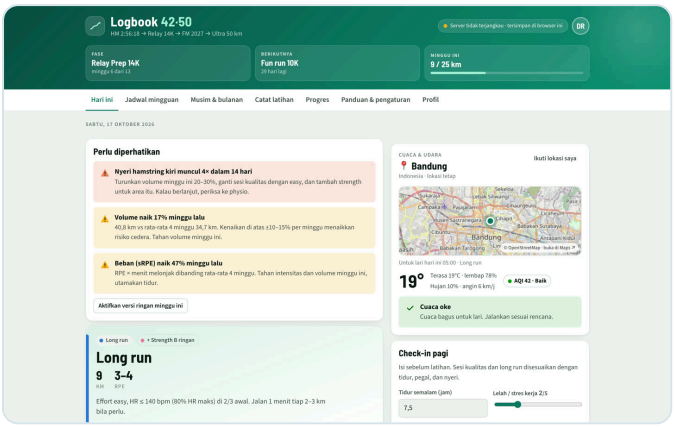
4. **intervals.icu instead of Strava**
   Huawei watches have no public API, and since June 2026 the Strava API requires a paid subscription. intervals.icu is free, connects to Huawei Health, and provides activities, streams (map, pace, HR), and wellness (VO2max, sleep, HRV, fitness/fatigue).

5. **Optional AI with fallbacks**
   Without GEMINI_API_KEY the app still works: screenshot import falls back to in-browser OCR. The Gemini caller tries several models, cuts off hung requests, stops before Vercel's 60-second limit, and tells the user when the daily quota runs out.

6. **Links derived from the password**
   The read-only coach link and the calendar feed use an HMAC token derived from APP_PASSWORD, since calendar apps can't send headers. Changing the password revokes every old link automatically, with no extra token table.

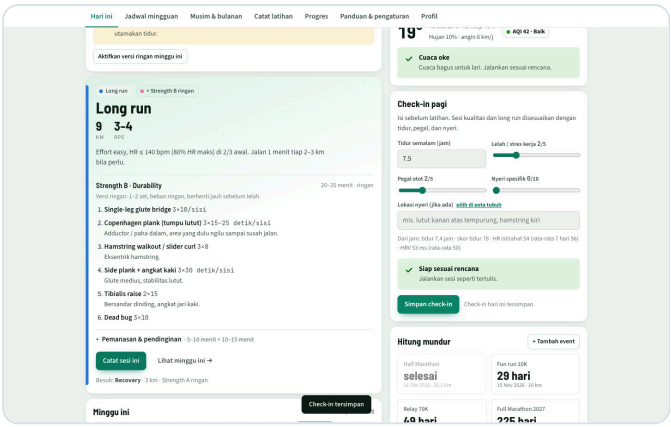
HOW TO USE IT

These steps follow the same order as the tutorial video.



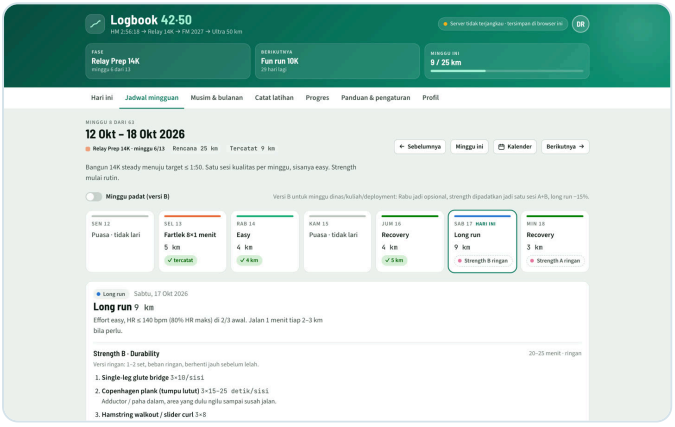
1 Open the Today tab

See today's session for the current phase, load warnings at the top, and the weather and air quality for your next run.



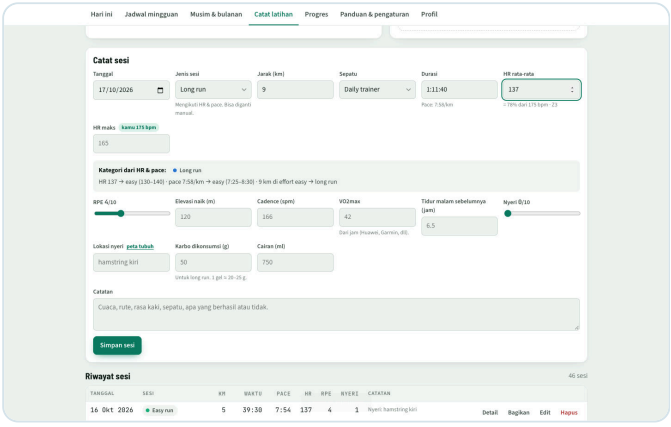
2 Fill in the morning check-in

Enter sleep hours, fatigue, soreness, and pain, then press Save check-in. The advice below updates immediately.



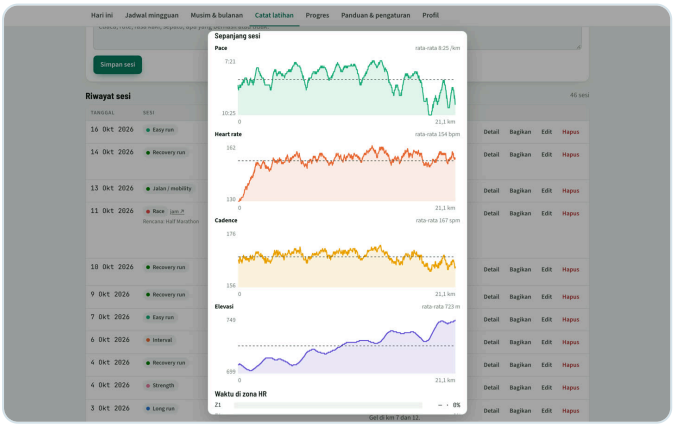
3 Check the weekly plan

The weekly plan tab shows each day's session, the week's km target, and which sessions are done.



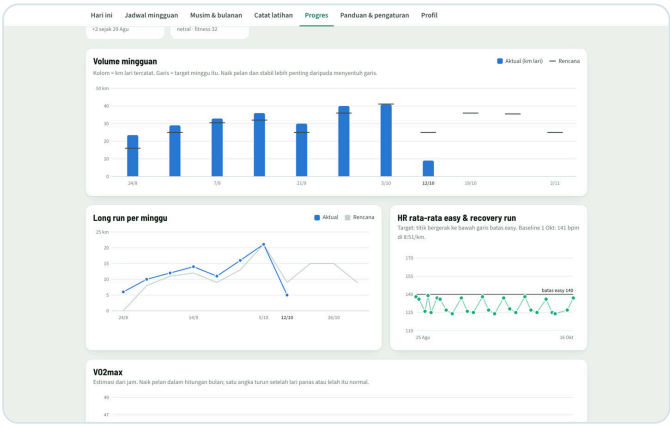
4 Log a session

In Log training, pick the session type and enter distance, duration, and HR. Pace and category are calculated for you; press Save session.



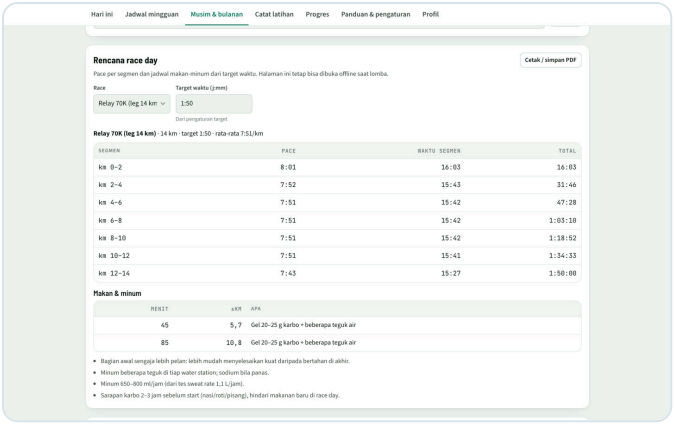
5 Open a session's details

Press Detail in the session history. Watch sessions show the map, pace/HR/cadence/elevation charts, HR zones, and per-km splits.



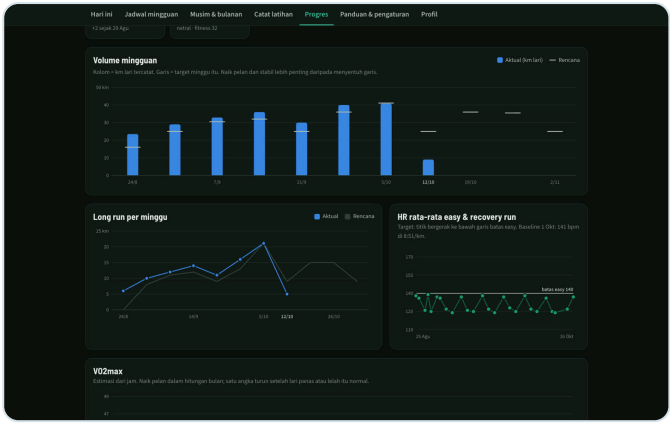
6 Track progress

The Progress tab has weekly volume, long runs, VO2max, Fitness · Fatigue · Form, achievements, and race predictions.



7 **Plan race day**

In Season & monthly, pick a race and a target time to get pace per segment and an eating and drinking schedule.



8 **Profile & theme**

The profile keeps personal records, HR zones, and the pain map. Light/dark theme is set under Display & notifications.

LIMITATIONS & NOTES

- One user per deployment: access is locked with a single password, with no account system.
- The planning and warning rules are personal training heuristics, not medical advice.
- The Fitness · Fatigue · Form chart, session detail with a map, and automatic check-ins need intervals.icu sync; manual sessions only show their stats.
- Gemini's free tier allows roughly 20 requests per day across reviews, questions, and imports.
- On the Hobby plan, Vercel Cron runs once a day within the given hour, so notification times are not exact.
- The app is private and password-protected, so there is no public demo; the portfolio video and screenshots use demo data in local mode.

WHAT'S NEXT

- A public demo mode with sample data and no password.
- Splitting index.html into modules with tests for the planning and warning logic.
- Multi-user support so running-club friends can use it too.